

Success Is Not Grounding: A Causal Audit of Language-Conditioned Manipulation Benchmarks

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Abstract—Directional failures in language-conditioned manipulation are commonly attributed to weak language grounding. We test this explanation using matched trials that hold the scene, initial state, sampling seed, checkpoint, and controller fixed while switching only the directional word between LEFT and RIGHT. Across 324 pairs spanning eight completed checkpoint–arena rows and two simulated environments, 253 endpoint pairs (78%) are ordered in the requested direction, showing that the instruction redirects behavior even when task success differs by direction. Scene geometry can determine whether this redirection becomes a successful placement. Reflecting movable-object positions reverses placement advantage in three checkpoints and flips the favored direction for $\pi_{0.5}$: LEFT/RIGHT completion changes from 4/27 and 25/27 in the original layout to 25/27 and 9/27 after reflection. Moving the reference object across seven lateral positions changes the placement gap gradually, while a symmetric scene reduces the gap across checkpoints that pass the endpoint-redirection control and brings DreamZero’s observed gap near zero. Canonical response, however, does not guarantee that the physical relation transfers across linguistic forms. For $\pi_{0.5}$, expressing the same goals from the other object’s frame collapses LEFT-minus-RIGHT endpoint separation from 23.3 to 0.2 cm. Binary directional success is therefore not a grounding test by itself: scene geometry can make language-responsive control look like failure, while familiar wording can make surface-form sensitivity look like semantic understanding.

Index Terms—Vision-language-action models, robot learning, manipulation, language grounding, evaluation methodology.

I. INTRODUCTION

Vision-language-action (VLA) policies are increasingly evaluated by asking a robot to perform the same task under counterfactual instructions. For relational placement, a natural test is to hold the scene fixed, replace *left* with *right*, and compare task success. A large LEFT/RIGHT gap is then tempting to read as a language-grounding failure: the model understood one direction but not the other, or vision overrode language. This interpretation has practical consequences. It directs effort toward stronger language conditioning, counterfactual relabeling, or steering at inference time [1]–[3].

The success gap, however, combines several questions. The instruction must alter control; the change must be oriented toward the requested relation; the robot must execute a feasible grasp and transport trajectory; and the final state must cross a binary task boundary. A policy can pass the first two steps but fail the last two because one side of a particular layout is harder to reach. The reverse mistake is also possible: a policy can react predictably to the literal tokens LEFT and RIGHT without preserving the same physical goal when the linguistic reference frame is inverted. Neither case is identified by a single binary contrast.

We audit this inference with two kinds of matched intervention. First, we compare canonical LEFT and RIGHT prompts under identical simulator and sampling state. Across 324 pairs, 253 endpoint pairs are ordered in the token-associated direction, while all 324 pairs execute different action sequences. Second, we express the same physical relation from the other object’s reference frame: “cube left of bowl” becomes “bowl right of cube,” and conversely. Under this reference inversion, $\pi_{0.5}$ does not preserve the canonical directional endpoint separation. Canonical token sensitivity is therefore not sufficient evidence of relation-level generalization.

Fig. 1 reports all eight completed checkpoint–arena rows before any diagnostic narrowing. Binary directional disparity varies sharply by checkpoint and arena. Later intervention figures therefore show their registered checkpoint subsets, and geometric equalization claims are restricted to rows that pass the relevant endpoint-redirection control. Within those subsets, we intervene on the scene while keeping prompts, policies, robot, cameras, controller, and seeds fixed. Position reflection reverses the placement advantage in three checkpoints; a seven-position displacement sweep grades it continuously; and a registered symmetric scene sharply reduces it across the qualifying DROID/RoboLab checkpoints.

Our contribution is an evaluation argument supported by controlled experiments:

- We show that canonical LEFT/RIGHT task success does not isolate language grounding. Action change, endpoint orientation, continuous goal progress, and binary completion must be measured separately.
- We introduce a reference-inversion control and show that canonical-token response need not transfer when the same relation is expressed from the other object’s frame.
- We provide matched reflection, graded displacement, role, start-side, and symmetric-scene interventions showing that scene configuration can reverse or largely remove a directional completion gap without changing the evaluated policy.

II. RELATED WORK

Generalist robot policies combine pretrained visual-language representations with expressive action decoders. Representative systems include π_0 and $\pi_{0.5}$, which use flow-matching action models, and OpenVLA, an open autoregressive alternative [4]–[6]. Their scale has expanded the tasks addressable by one policy, but it also makes a failed episode difficult to attribute: language interpretation, object localization, grasping, transport, release, and task scoring all lie on the same path to success.

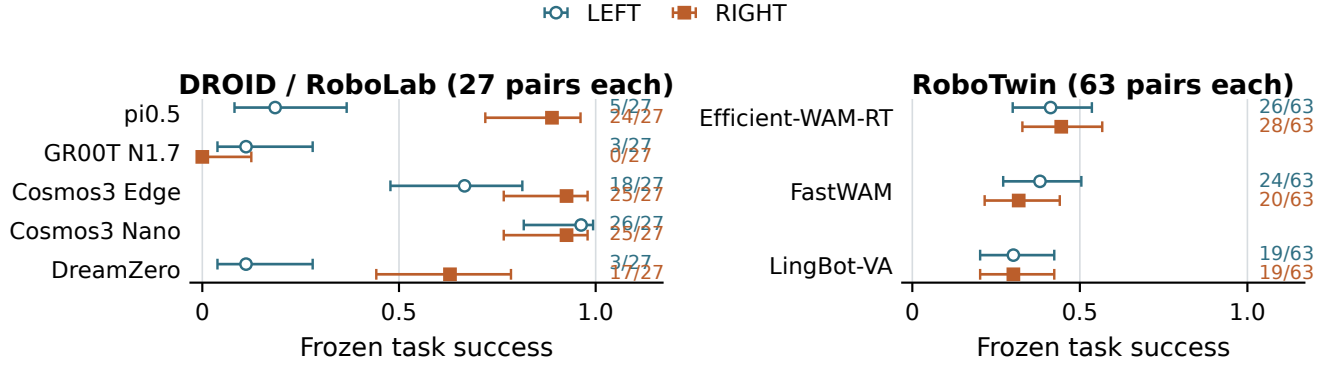


Fig. 1. **Directional task success varies across all eight completed checkpoint-arena rows.** Points show marginal LEFT and RIGHT success with 95% Wilson intervals. DROID/RoboLab uses 27 matched pairs per row and RoboTwin 63; their tasks and frozen predicates differ and are never pooled. This screen is separate from later intervention cohorts, and its intervals are descriptive rather than matched-pair tests.

Counterfactual evaluations reduce this ambiguity by holding much of the visual input fixed while changing the instruction. LIBERO-CF and Counterfactual Action Guidance examine cases in which visual evidence dominates a conflicting instruction [1]; CAST expands instruction diversity through counterfactual labels [2]; and Steerable Policies train controllers on task, motion, trace, and point commands [3]. These methods address genuine failures of instruction following. Our question is complementary: when the objects are already fixed and the requested relation changes, does a LEFT/RIGHT completion contrast uniquely diagnose language grounding? Target contact, for example, can verify object selection but cannot verify whether the target was placed on the requested side of the reference.

Manipulation benchmarks typically retain a frozen binary task predicate, as they should. DROID and RoboTwin illustrate the diverse data and simulation ecosystems used to train and test modern policies [7], [8]. A binary predicate answers whether the task was completed in the evaluated scene distribution. It can nevertheless saturate at ceiling, collapse at floor, or hide large continuous placement changes. We keep the original predicate as the final outcome, but pair it with signed endpoint and requested-depth measurements and with interventions that vary language form and scene geometry separately.

III. EVALUATION DESIGN

Canonical token intervention. For each registered reset s_i , a policy is executed under prompts ℓ_i^L and ℓ_i^R that differ only in the relation token. Scene state, checkpoint, controller, environment seed, policy-sampling seed, action interface, and horizon are held fixed. The instruction is issued once at reset; oracle actions, prompt switching, and progress-conditioned language are prohibited. DROID/RoboLab uses a Rubik’s cube and bowl, while RoboTwin uses a wooden block and playing-card box. Both require placement and release in the requested relational region. The canonical screen contains five DROID/RoboLab checkpoints with 27 matched pairs each and three RoboTwin checkpoints with 63 matched pairs each, for

324 pairs total; arena-specific tasks and predicates are never pooled.

Reference inversion. The canonical contrast changes the surface token and physical goal together. C002 instead holds the physical relation fixed while changing its linguistic frame: “cube left of bowl” becomes “bowl right of cube,” and conversely. Thus the direction word flips but the intended relation does not. The registered $\pi_{0.5}$ study uses 341 reset blocks in one symmetric DROID/RoboLab scene, with both forms of both physical goals sharing reset and sampling state, for 1,364 episodes. An initial execution failed isolation and is excluded; the prospective rerun uses exact within-server repeatability and block-local homogeneous lanes. Its seeds 12000–12340 are independent of the E004 cohort at 9400–9740.

Behavioral measurements. Let $a_{1:T_L}^L$ and $a_{1:T_R}^R$ be the executed action sequences in a canonical pair, and let y_i^L, y_i^R be the final target-minus-reference lateral offsets in robot coordinates, where positive is robot-left. We report

$$A_i = \mathbb{K}[a_{1:T}^L \neq a_{1:T}^R], \quad T = \min(T_L, T_R), \quad (1)$$

$$D_i = y_i^L - y_i^R, \quad (2)$$

$$S_i^d = \mathbb{K}[\text{frozen task predicate is satisfied for } d], \quad (3)$$

$$M_i^d = \begin{cases} y_i^d, & d = L, \\ -y_i^d, & d = R. \end{cases} \quad (4)$$

A_i uses bitwise inequality on the complete common executed prefix. Positive D_i means the canonical endpoints have the requested ordering. S_i^d is task completion, and requested-side depth M_i^d retains how far the target moved toward the physical goal. For reference inversion, M is signed by the physical goal rather than the word appearing in the prompt. Action distinctness tests response, not semantics: stochastic executions can differ even under an unchanged prompt.

Geometry interventions. We use three registered scene interventions. Reflection mirrors the centers of all movable objects about the robot sagittal plane while leaving the robot

and fixed scene unchanged. A seven-level sweep moves only the reference object laterally. The symmetric-scene cohort sets the registered object-layout residual to zero while preserving robot, cameras, wrist mounting, reset posture, prompt, seed, and controller. Its registered scene package also changes companion-object inventory, so it is not a position-only intervention and does not imply bilateral embodiment symmetry. Each causal scene comparison requires the canonical endpoint-redirection positive control to pass; rows failing it are reported but excluded from equalization claims.

Scoring and inference. The original task predicate is never relaxed. Each valid episode receives one precedence class: pick failed, transport failed, wrong side, release failed, or correct. Behavioral failures remain in their registered denominators; infrastructure-invalid attempts are ledgered separately. Marginal binary rates use Wilson intervals. Paired continuous contrasts use seed-clustered percentile bootstraps with 20,000 resamples and exact sign tests; layout interactions use exact within-seed label permutations. Absence claims require two one-sided equivalence tests (TOST) against preregistered margins, rather than nonsignificance [9]–[11]. Prompts, seeds, predicates, estimands, margins, and failure precedence are frozen before their affected runs.

IV. CANONICAL RESPONSE IS NOT SEMANTIC EQUIVALENCE

A. Canonical LEFT/RIGHT response

All 324 canonical LEFT/RIGHT pairs across the eight completed rows execute different action sequences: 135 DROID/RoboLab pairs and 189 RoboTwin pairs. Endpoint ordering is positive in 125/135 and 128/189 pairs, respectively. These observations reject the narrow hypothesis that changing the canonical direction token leaves control unchanged. They do not show that the policy represents the relation independently of the wording, nor do they guarantee task completion.

A fixed-observation diagnostic issued 336 action requests without executing the robot; all 12 exact repeats were bit-identical. For $\pi_{0.5}$, prompt-induced action difference was about 4% of cross-seed same-prompt variation but aligned across seeds. Cosmos3 Nano showed a larger, less consistently oriented effect, while DreamZero had no same-prompt variation. These open-loop rankings did not match closed-loop endpoint ordering, so action difference is a positive control, not a standalone grounding score.

B. Reference inversion

C002 asks whether the same physical relation is preserved when expressed from the other object’s frame. Under inversion, the surface direction word and physical goal point in opposite directions. Relation preservation therefore predicts a positive physical-LEFT-minus-physical-RIGHT endpoint contrast. A simple template-invariant direction-token rule instead predicts a negative contrast—23.28 cm if the canonical response magnitude transferred unchanged.

Fig. 2 shows that neither idealized response occurs. Canonical prompts separate the endpoints by +23.28 cm (95% CI

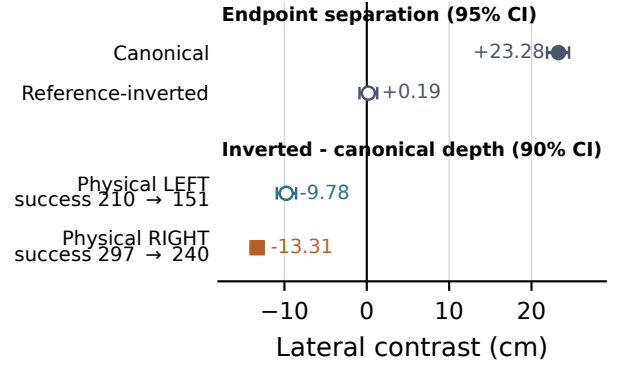


Fig. 2. **Reference inversion collapses directional endpoint separation for $\pi_{0.5}$.** Top: physical-LEFT-minus-physical-RIGHT endpoint separation with seed-clustered 95% bootstrap intervals; positive is relation-consistent. Bottom: inverted-minus-canonical requested-side depth with 90% intervals; labels give canonical-to-inverted success counts out of 341. C002 seeds 12000–12340 are independent of E004.

[21.89, 24.56]), whereas reference-inverted prompts yield only +0.19 cm ([−0.88, 1.27]). The directional endpoint response therefore collapses under inversion: it provides neither reliable relation-consistent ordering nor evidence for simple canonical-scale token mirroring. This conclusion is specific to endpoint direction, not all prompt sensitivity; all 682 canonical/inverted physical-goal pairs execute different action traces.

Within each physical goal, inversion also reduces placement depth. The inverse-minus-canonical change is −9.78 cm for physical LEFT (90% CI [−10.92, −8.59]) and −13.31 cm for physical RIGHT ([−14.07, −12.54]); both intervals lie outside the registered ± 4.15 cm equivalence margin. Success falls from 210/341 to 151/341 for LEFT and from 297/341 to 240/341 for RIGHT. The corresponding changes, −17.30 and −16.72 percentage points, do not meet the registered ± 15.56 -point equivalence criterion. The inverse endpoint-ordering control also fails, so the semantic-equivalence claim is withheld. Canonical response is therefore insufficient evidence that a physical relation is preserved across reference frames.

V. DIRECTIONAL COMPLETION TRACKS OBJECT GEOMETRY

The complete Phase-A screen in Fig. 1 contains five DROID/RoboLab and three RoboTwin rows. Large RIGHT-over-LEFT marginal gaps appear for $\pi_{0.5}$ and DreamZero. GR00T is floor-limited at 3/27 LEFT versus 0/27 RIGHT and is the only row whose small observed difference favors LEFT; Nano is near ceiling, and the RoboTwin gaps are small. Later experiments use registered checkpoint subsets, with geometric claims further restricted by endpoint-redirection controls. The screen is neither an architecture ranking nor a pooled cross-arena estimate.

A. Position reflection

We hold the robot, cameras, fixed geometry, prompts, checkpoints, and seeds constant while reflecting the centers of movable objects about the sagittal plane. Each of three

DROID/RoboLab checkpoints uses 27 seeds and four cells (two directions by two layouts), for 108 episodes. The $\pi_{0.5}$ reflection cohort uses seeds 9400–9426, independently of the Phase-A screen at 8303–8329. Thus Fig. 1 reports 5/27 LEFT and 24/27 RIGHT, whereas the control cell below contains 4/27 and 25/27. The primary outcome is the within-seed layout interaction in requested-side depth; success is secondary because binary outcomes can saturate by layout and direction.

As Fig. 3 shows, requested-side depth reverses in every checkpoint: -34.6 cm for $\pi_{0.5}$ (95% CI $[-41.4, -28.5]$; 27/27 seeds; $p = 1.49 \times 10^{-8}$), -24.8 cm for Nano ($[-32.4, -17.3]$; 24/27; $p = 4.92 \times 10^{-5}$), and -14.1 cm for DreamZero ($[-19.9, -8.2]$; 23/27; $p = 3.11 \times 10^{-4}$). For Nano and $\pi_{0.5}$, the estimated interactions in canonical endpoint redirection are much smaller, $+0.5$ cm and -1.12 cm (both $p = .701$). Because equivalence was not established for those estimates, we do not call redirection invariant; the supported contrast is that placement changes far more than the estimated redirection interaction.

For $\pi_{0.5}$, binary completion itself inverts. The control layout yields 4/27 LEFT versus 25/27 RIGHT successes; reflection yields 25/27 versus 9/27, a difference-in-differences of -1.37 ($p = 5.96 \times 10^{-8}$). The checkpoint, prompts, seeds, controller, and robot are unchanged. The benchmark’s favored direction changes when the movable objects change sides.

B. A graded intervention and converging controls

Reflection is a large intervention, so we also move the bowl across seven registered lateral positions while retaining the same Nano checkpoint and 15 matched seeds per level (210 episodes). The requested-depth contrast changes gradually over the full support, with a slope of 1.12 m/m (95% CI $[0.72, 1.56]$); 13 of 15 seed-level slopes are positive (exact $p = .00739$). A post hoc middle-five sensitivity is weaker: 0.49 m/m (95% CI $[-0.04, 0.97]$), with 11 of 15 positive slopes ($p = .118$). We therefore claim a graded dependence across the registered seven positions, not an interior linear law. The fitted zero crossing lies outside the tested support.

Fig. 4 also shows why binary success alone is a weak diagnostic here: LEFT completion changes from 15/15 to 10/15, while RIGHT stays close to ceiling. Separate interventions point in the same direction. Moving the target’s start side changes success from 26/27 versus 22/27 to 23/27 versus 27/27 (interaction $+0.296$, $p = .0156$). Swapping target and reference roles changes the depth contrast by $+0.13$ m (95% CI $[0.06, 0.21]$, $p = .00151$). Finally, 16 policy samples per fixed scene yield 41/216 LEFT and 197/216 RIGHT successes. The gap persists when policy sampling is repeated in these scenes; it is not an artifact of one draw.

VI. A SYMMETRIC SCENE REDUCES THE GAP

The registered symmetric-scene cohort contains 4,096 valid episodes and 2,048 matched pairs across five checkpoint–arena rows. The intervention removes the registered object–layout residual while preserving the robot, camera rig, wrist mounting, reset posture, prompt, seed, and controller. It also

uses a registered companion-object inventory transition and is therefore a scene-package intervention, not a position-only edit. Four DROID/RoboLab checkpoints pass the canonical endpoint–redirection positive control and support this comparison. The RoboTwin FastWAM row fails closed and is not used for cross-arena equalization claims. Infrastructure-invalid attempts remain outside behavioral denominators. The powered $\pi_{0.5}$ endpoint uses seeds 9400–9740 and is independent of C002, whose 341 blocks use seeds 12000–12340.

For $\pi_{0.5}$, the full 341-seed endpoints show the binary RIGHT-minus-LEFT gap falling from 74.2 to 20.2 percentage points and the requested-depth gap from 17.0 to 5.75 cm. At the symmetric endpoint, their paired 90% intervals are $[15.0, 25.2]$ points against a registered ± 15.56 -point margin and $[4.85, 6.66]$ cm against ± 4.15 cm; neither is equivalent. The separately defined matched 27-seed core estimates interactions of -51.9 points (95% CI $[-81.5, -22.2]$, $p = .00786$) and -12.4 cm (95% CI $[-19.7, -4.9]$, $p = .00348$), respectively. Canonical redirection at the symmetric endpoint is $+23.9$ cm, with 335/341 positive pairs. The larger sample estimates that endpoint, while the matched core supplies the scene-intervention test.

The matched core also shows why no single summary suffices. DreamZero’s observed binary gap changes from $+51.9$ to -3.7 points (interaction -55.6 points, $p = .00085$), bringing the measured gap near zero without establishing equivalence. Nano has exactly zero binary interaction at ceiling ($p = 1$), yet its 27-seed depth interaction is -13.9 cm ($p = 1.8 \times 10^{-4}$). Cosmos3 Edge is underpowered for its -14.8 -point binary interaction, while its depth interaction is -19.2 cm (95% CI $[-26.0, -11.5]$). Fig. 5 keeps these estimands separate.

The same intervention changes failure composition (Fig. 6). Transport failures fall from 267 to 158, wrong-side failures from 41 to 7, pick failures from 1 to 0, and release failures from 2 to 0. Thus the scene package improves manipulation outcomes without changing the policy or prompt.

VII. DISCUSSION

The benchmark can be wrong in both directions. The experiments expose two distinct evaluation errors. A *false failure* occurs when a canonical direction token redirects behavior, but the sampled scene makes one requested placement harder to complete. Reflection, displacement, start-side, role-swap, and the symmetric scene all demonstrate this possibility. A *false pass* occurs when relation-consistent behavior under a familiar canonical construction is treated as evidence of grounding even though the same physical goal does not transfer under reference inversion. In C002, relation preservation and simple canonical-scale token mirroring predict opposite-signed endpoint contrasts, yet the observed inverted contrast is near zero. Together, these interventions show that a canonical success score can understate language-conditioned control or overstate semantic transfer.

Success remains necessary, but not diagnostic. The frozen success predicate is still the correct final test of whether

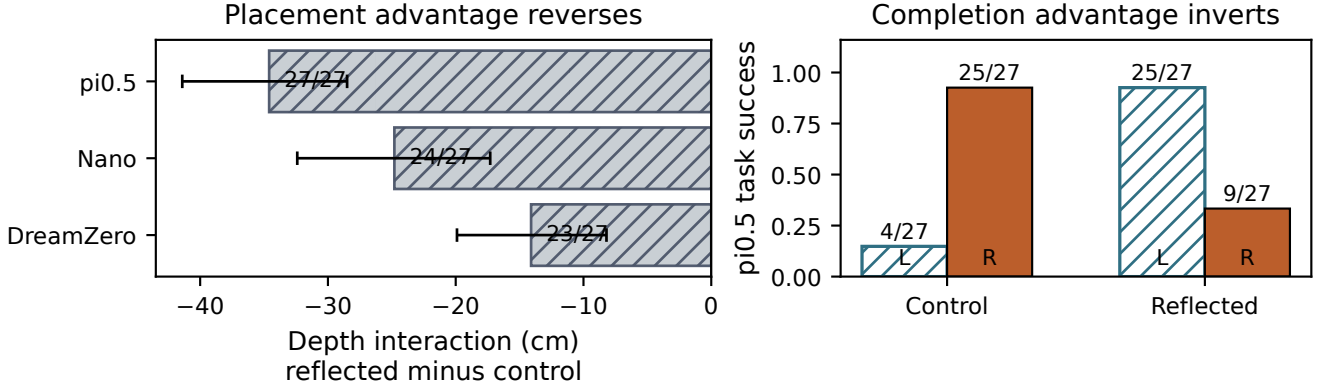


Fig. 3. **Reflecting movable-object positions reverses placement advantage.** Left: reflected-minus-control interaction in requested-side depth, with seed-clustered 95% bootstrap intervals; fractions give seeds with the negative interaction. Right: $\pi_{0.5}$ task completion in the independent reflection cohort. Nano is near a binary ceiling, while DreamZero changes from low control completion to near-ceiling reflected completion; their continuous depth interactions remain the common estimand.

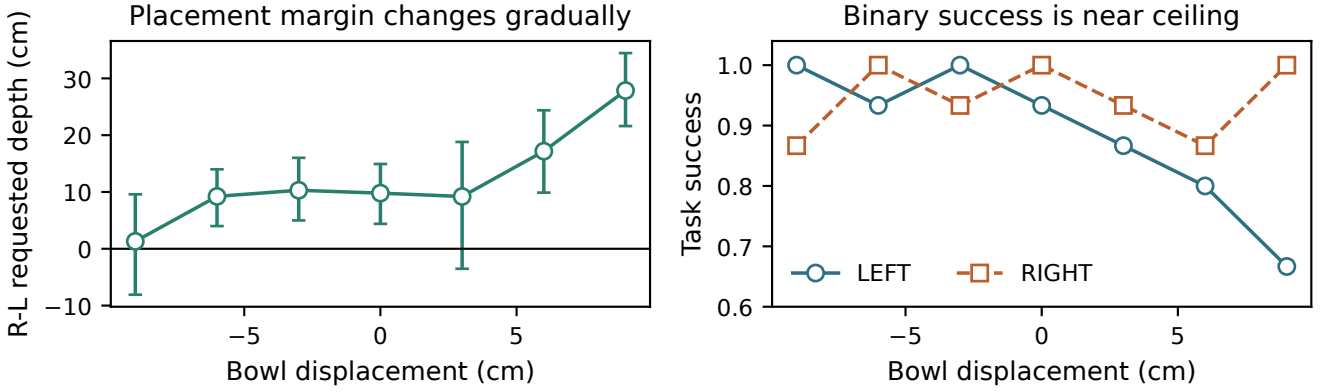


Fig. 4. **The directional placement contrast changes across a seven-position reference-object sweep.** Left: mean RIGHT-minus-LEFT requested-side depth with seed-clustered 95% intervals. Right: frozen task success by requested direction. Each position contains 15 matched seeds; the continuous measure exposes variation that binary success compresses near ceiling.

the robot completed the task. The problem is using that predicate alone to infer *why* it succeeded or failed. A single layout can make scene geometry and language interpretation observationally indistinguishable. A ceiling can also hide a 10–20 cm change in placement quality. Conversely, an action trace can change without expressing the intended relation. The four behavioral measurements answer different questions, and the reference-inversion control adds a fifth: whether the response follows the physical goal rather than the surface token.

A minimum directional-grounding protocol. For relational placement, we recommend: (i) matched canonical LEFT/RIGHT prompts under identical reset and policy seeds; (ii) action distinctness and a direction-signed endpoint measure; (iii) continuous requested-side depth alongside the frozen success predicate; (iv) same-prompt repeats where stochasticity is material; (v) at least one matched scene intervention with a preserved endpoint-redirection positive control; and (vi) a reference inversion that preserves the physical goal. Claims that an effect disappeared should use equivalence tests, not a

nonsignificant difference.

VIII. LIMITATIONS

First, all behavior is simulated. Hardware contact uncertainty and natural variation in human instructions remain untested. The causal object-layout conclusions are restricted to the positive-control-passing DROID/RoboLab rows; RoboTwin screening results cannot be read as cross-arena replication or failure.

Second, the task family contains two named objects, one static instruction, and one lateral relation. C002 tests one controlled reference inversion for one $\pi_{0.5}$ checkpoint in one symmetric scene. It is not a broad paraphrase benchmark and does not establish unrestricted compositional grounding.

Third, reflection moves object centers, whereas the symmetric condition changes a registered scene package that also includes companion-object inventory. Both leave embodiment, camera viewpoint, wrist mounting, action parameterization, and decoder unchanged. They identify dependence of the measured outcome on scene configuration; they do not localize errors inside the model or robot system. Checkpoint provenance is

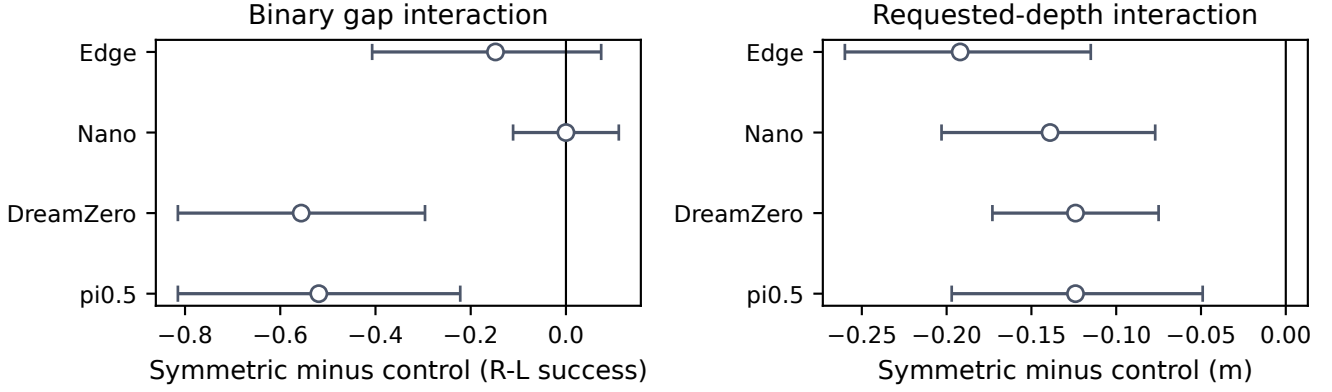


Fig. 5. **The registered symmetric scene reduces directional disparity in the positive-control-passing DROID/RoboLab cohort.** Points show symmetric-minus-control interactions in the matched 27-seed core, with seed-clustered 95% intervals. Requested-depth disparity decreases in every checkpoint. Binary interaction is compressed where the control policy is already at ceiling, as for Nano.

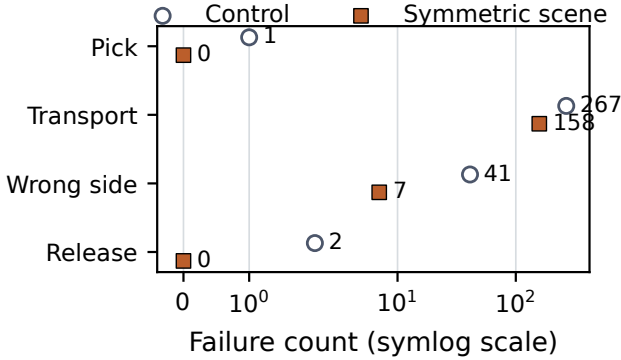


Fig. 6. $\pi_{0.5}$ **failure composition shifts in the matched E004 cohort.** Counts pool directions over 341 matched seed pairs (682 episodes per scene). All registered failure categories, including release, are retained. The plot is descriptive; it does not localize an internal policy mechanism.

also not fully auditable for every evaluated model, so we avoid architecture-level attribution.

Finally, the fixed-observation diagnostic is action-space only and does not bind prompt-conditioned actions to verified forward kinematics. These limits narrow the causal interpretation, but they do not alter the central evaluation result: the same binary directional score can change under reference inversion or scene interventions.

IX. CONCLUSION

Binary LEFT/RIGHT task success does not, by itself, measure language grounding. In our matched trials, canonical direction tokens redirect control, but scene configuration can reverse which direction completes successfully, and reference inversion can remove the expected response. Directional benchmarks should therefore report prompt-conditioned action change, signed endpoint ordering, continuous placement, binary completion, scene controls, and semantic-form controls separately before assigning a failure to weak language grounding.

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